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OSI Model vs TCP/IP Model

1. Introduction

The OSI model is a seven-layer reference model used to understand network communication. The TCP/IP model describes the practical protocol architecture used by the internet.

The OSI model is useful for structured troubleshooting, while TCP/IP helps explain how real-world network communication works.

2. OSI Layers, Protocols and Illustrative Security Threats

Table 1

Layer	Name	Examples	Illustrative threat or attack
7	Application	HTTP, HTTPS, DNS, SMTP	SQL injection against a vulnerable web application
6	Presentation	TLS encoding, data formats	Exploitation of weak encryption or unsafe data handling
5	Session	Session management, RPC	Session hijacking
4	Transport	TCP, UDP	TCP SYN flood
3	Network	IPv4, IPv6, ICMP	IP spoofing
2	Data Link	Ethernet, ARP, 802.1Q	ARP spoofing
1	Physical	Ethernet cabling, radio signals	Physical cable tapping or deliberate disconnection

These are examples of threats associated with each layer, not an exhaustive classification. Some attacks span multiple layers.

3. TCP/IP Model Mapping

TCP/IP layer	Approximate OSI mapping	Examples
Application	OSI 7, 6 and 5	HTTP, DNS, SMTP, SSH
Transport	OSI 4	TCP, UDP
Internet	OSI 3	IP, ICMP
Network Access / Link	OSI 2 and 1	Ethernet, Wi-Fi, ARP

Table 2

Terminology and the exact number of TCP/IP layers vary across textbooks. The four-layer model is used here.

4. Example of Network Traffic

When a user opens an HTTPS website:

1. The browser prepares an application request.
2. Transport protocols carry the data.
3. IP addresses help route packets to the destination.
4. Ethernet or Wi-Fi transmits frames over the local link.
5. The receiving device processes the data through its networking stack.

5. Why SOC Analysts Need This Knowledge

- Identify which layer may be responsible for a network problem.
- Interpret firewall, DNS, endpoint and packet-capture logs.
- Understand the difference between network and application attacks.
- Correlate activity across multiple security controls.

6. Conclusion

Both models help explain communication between systems. The OSI model provides detailed conceptual separation, while TCP/IP reflects the architecture commonly used in operational networks.